

Samples.csv = metadata file

- Put this into module 4 folder

Promicromonospora.gff - annotated genome

Promicromonospora.fasta - genome in a list

Step 1.) check for quality with fastqc (use falco [Galaxy Version 1.2.4+galaxy](#) because its faster)

- Default parameters fine
- Download forward and reverse quality reads
 - Forward data: fails in Per Tile Sequence Quality, PerBase Sequence Content, Per Sequence GC Content, Sequence Duplication Levels, and Overrepresented Sequences
 - Warning in the Sequence Length Distribution
 - Reverse Data: same fails and warning as forward sequence reads

Step 2.) trim with fastp ([Galaxy Version 1.3.1+galaxy0](#))

- Set to paired collection
- Gets rid of stuff that are fails for the quality control
 - Observe the results on HTML, those under filtering reports

Step 3.) HISAT2 ([Galaxy Version 2.2.2+galaxy0](#)) - aligns transcriptome reads to the genome that is provided by Dr. Van Laar

- Change in parameters:
 - Source for reference genome: genome from the history
 - Single or paired library? Set to paired-end dataset collection

Made an alignment file called BAM

Now, what actual transcripts do we have?

Step4.) StringTie ([Galaxy Version 2.2.3+galaxy0](#)) - assembles short reads into actual transcripts, predicts full length transcripts

- Make sure input option parameter is set to Short Reads

Created 6 files of transcripts

- Change parameter to reference genome to history
 - promicromonospora

Step 4.) StringTieMerge ([Galaxy Version 2.2.3+galaxy0](#)) - instead of 6 files, puts them all together

- Use dataset collection and use the StringTie file
- Use reference as promicromonospora

-new stuff here dont do gffcompare

- step 5.) featureCounts

- Set alignment file to dataset collection and use HISAT file
- Set gene annotation file as GFF/GTF from history and use latest stringtiemerge
- Rewrite GFF gene identifier as transcript_id
- Does the input have read pairs? Set to Yes, paired-end and count them as 1 single fragment

- Deseq2 (Galaxy Version 2.11.40.8+galaxy2) - function? Takes all 6 conditions and sees if it is more represented in one condition or the other, gives statistical significance
- Factor: set to Growth_Media
- 2 factor levels. Factor level 1 set as Broth; Factor level 2 set as Plate
 - Tells if gene was up or down regulated
 - Use beta priors? Set to yes
 - Output options: generate plots and output normalised counts
 - Click “include hidden” within history and locate the featurecount and find the Counts, rename as plate/BrothReplicate#
 - Put these in using the multiple dataset options, search with the ellipses
- From the result file you get a p-value, can determine null hypothesis
 - For the gene, there is no change in gene expression between plate and broth: null hypothesis
 - If there are changes between the two growth media, you can infer significant difference, but need this p-value to make a statistical backing for your reasoning
 - Less than 0.05, it is statistically significant

Step 6.) filter data on any column using simple expressions

- Use on DEseq2 file
- With following condition - $c7 < 0.05$
- Run again with the following condition - $\text{abs}(c3) > 1$
 - Only do this after the previous filtering is finished
- From the plots we download here, we learn about whether expression change between the two growth media/conditions is statistically significant enough to reject the null hypothesis

Step 7.) filter on dataset with following condition:

- $\text{abs}(c3) > 1$
 - Cut it down to 1,308 lines (still a lot but a lot less than 4,000 +)

Step 8.) were gonna add a header now through a new textedit file

- Put the following in: GeneID Base mean $\log_2(\text{FC})$ StdErr Wald-Stats P-value P-adj

Step 9.) concatenate tool

- Insert the most recent filter on dataset file

Now our data file has proper headers!

- Rename the file to Differential Gene Expression of Broth and Plate

Step 9.) volcano plot (Galaxy Version 4.0.2+galaxy0) - represents p-values

- Change the respective columns into the following: 7, 6, 3, 1

Step 10.) opened the tabular broth vs plate file as an excel file

- Sort the entire document from smallest to largest in $\log_2(\text{FC})$, kept the top 10 and bottom 10, delete everything in between, then save as a .csv file (separately so that you can still have the initial file)
- Make sure you delete all the cells that aren't included in the list

Step 11.) upload that new.csv file to galaxy

- Edit attribute on galaxy > datatypes > make sure New Type is set as “tabular”

Step 12.) join 2 datasets side by side on a specified field (Galaxy Version 2.1.3)

- Join .csv file with normalized counts dataset
- Parameter change: Keep Header Lines - Yes

Step 13.) cut columns from a table

- parameter : cut c8-c14
 - *may need to set as c9-c14 if there is a blank column, check by clicking on the file in galaxy and scrolling to the right to see if there is an additional column*

Step 14.) heatmap2 -

- Change parameters:
 - Data transformation - Log2(value+1) transform my data
 - Blue (high level expression)
 - Brown (low level of expression)

Step 15.) filter data on any column using simple expressions

- Filter - promicromonospora.gff
- Wanna keep all under CDS under column 3
 - With following condition: c3=='CDS'

Now we wanna keep column 9

Step 16.) Cut column from a table

- Retain only column 9
- Parameters: cut columns - c9

Cant read these intuitively so...

Step 17.) convert delimiters to TAB

- Parameters:
 - Convert all - semicolons

We just want the gene name on column 2

Step 18.) replace text in a specific column

- Parameters:
 - In column 2
 - Find pattern - Parent=
- Can preview under visualize for confirmation

Now we're interested in column 8 because it has the gene function
We want the gene name column 2 and the gene products column 8

Step 19.) cut columns from a table

- Parameters:

- Cut columns: c2,c8

Step 20.) join two datasets side by side on a specified

- Join cut on dataset with .csv file
- Keep header lines - yes